

## **PW1: Object-relational databases**

### **GRADING CRITERIA**

**Only files submitted to ORTUS by the specified submission deadline will be evaluated. No other additions (via email, comments, or other means) will be considered, regardless of the submitter's explanations.**

**Distribution of the overall score (200 points) across assessment units**

| Assessment unit                            | Project Passport (PP) | Code       | Extended ER Diagram | Class Diagram | Total (pts) |
|--------------------------------------------|-----------------------|------------|---------------------|---------------|-------------|
| <b>Tables</b>                              | <b>12</b>             | <b>72</b>  | <b>24</b>           | <b>12</b>     | <b>120</b>  |
| • Tables with row type objects             | 4                     | 24         | 8                   | 4             | 40          |
| • Tables with object columns               | 4                     | 24         | 8                   | 4             | 40          |
| • Tables with object collections           | 4                     | 24         | 8                   | 4             | 40          |
| <b>Object methods</b>                      | <b>4</b>              | <b>24</b>  | <b>10</b>           | <b>2</b>      | <b>40</b>   |
| <b>Functions &amp; operators</b>           | <b>2</b>              | <b>18</b>  | —                   | —             | <b>20</b>   |
| <b>Inheritance &amp; method overriding</b> | <b>2</b>              | <b>6</b>   | <b>6</b>            | <b>6</b>      | <b>20</b>   |
| <b>Total (points)</b>                      | <b>20</b>             | <b>120</b> | <b>40</b>           | <b>20</b>     | <b>200</b>  |

The work is graded as **0 points** without further evaluation in any of the following cases:

- 1) If the report is not submitted, or the submitted file is not a valid PDF file, or it is corrupted.
- 2) If the code file is not submitted, or the submitted file is not a valid TXT file, or it is corrupted.
- 3) If the work is not developed according to the domain description assigned to the group (e.g., if it is developed for a different domain or largely disregards the rules outlined in the domain description).

The Project Passport and Code are not evaluated at all (score is **0 points**) in any of the following cases:

- 1) If the report does not contain a valid link to the LiveSQL code.
- 2) If the code file submitted to ORTUS differs from the code downloaded during the evaluation from the LiveSQL link by more than just comments or similar details (e.g., even sequence differences).
- 3) If the Project Passport is not filled out.

**If the Project Passport and Code are not evaluated, only the diagrams are assessed to determine whether they qualify as implementation-level diagrams, with a maximum of 50% of the possible points for diagrams awarded. Otherwise, the evaluation continues with the next step.**

General rules for evaluating the Project Passport, Code, and related diagrams:

- 1) Code is evaluated only to the extent that it is referenced in the Project Passport.
- 2) If the LiveSQL code contains other incorrect code (not referenced in the Project Passport), this is also taken into account, and the score is reduced accordingly.
- 3) Before evaluation, the code is **re-executed**, and then the evaluation is performed.
- 4) The LiveSQL code must have visible outputs confirming successful execution – only those **statements with visible confirmation of successful execution are evaluated**.
- 5) Diagrams are evaluated together with the corresponding code.
- 6) If the diagrams are unreadable (e.g., font size smaller than Arial 10 pt (compared to the main text), blurred text), **the diagram is considered non-existent**.

## Table Evaluation

Each table type is evaluated separately:

- 1) Tables with row type objects
- 2) Tables with object columns
- 3) Tables with object collections

Remember, a single table can belong to multiple types!

### Evaluation for Each Table Type

| # | Criteria                                                                                                                                                                                                                                                                                                                                        | Max. Points |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 1 | <ul style="list-style-type: none"><li>• Table names in PP match precisely with LiveSQL code</li><li>• Statement numbers are correct</li><li>• <b>If neither the Statement number nor the table name is correct, further evaluation is not performed</b></li><li>• Tables are listed in the correct PP section (with the correct type)</li></ul> | 4           |

| # | Criteria                                                                                                                                                                                                                                                                                                                                                                                                                     | Max. Points |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 2 | <ul style="list-style-type: none"> <li>Successful creation of the table itself (CREATE) and related types demonstrated – 5 pts</li> <li>All code statements include meaningful comments – 2 pts</li> <li><u>For tables with object columns</u> – if none of these tables have REF columns: <b>-2 points (minus 2 pts)</b></li> </ul>                                                                                         | 7           |
| 3 | <ul style="list-style-type: none"> <li>Successful data insertion demonstrated (at least a few rows showing several possible, essentially different situations – many identical rows will not improve the score) – 6 pts</li> <li>All code statements include meaningful comments – 1 pt</li> <li><u>For tables with object columns</u> – if none of these tables have REF columns: <b>-3 points (minus 3 pts)</b></li> </ul> | 7           |
| 4 | <ul style="list-style-type: none"> <li>Successful data retrieval demonstrated (detailed data) – 6 pts</li> <li>All code statements include meaningful comments – 1 pt</li> <li><u>For tables with object columns</u> – if none of these tables have REF columns: <b>-3 points (minus 3 pts)</b></li> </ul>                                                                                                                   | 7           |
| 5 | <ul style="list-style-type: none"> <li>The rule is followed that immediately after the type is created (as soon as possible), table creation, data insertion, and retrieval are demonstrated (points are awarded if the code works)</li> </ul>                                                                                                                                                                               | 3           |
| 6 | <ul style="list-style-type: none"> <li>Tables are correctly reflected in the ER diagram (all data match, notation is correct and properly applied, etc.)</li> </ul>                                                                                                                                                                                                                                                          | 8           |
| 7 | <ul style="list-style-type: none"> <li>Tables are correctly reflected in the class diagram (all data match, notation is correct and properly applied, etc.)</li> </ul>                                                                                                                                                                                                                                                       | 4           |
|   | <b>Total:</b>                                                                                                                                                                                                                                                                                                                                                                                                                | <b>40</b>   |

**A meaningful comment** explains the purpose, logic, or complex solutions of the code, as well as highlights constraints or assumptions that are not immediately apparent from the code. **A comment is not meaningful** if it merely restates obvious code or provides no additional value to the reader.

**All types and tables must comply with the domain description (no contradictions are allowed)!**

## Evaluation of Object Methods

The following criteria are evaluated collectively for all methods:

### 1) Project Passport (max. 4 pts):

- PP includes correct statement numbers for method declaration, definition, and testing.
- Method names in PP precisely match the code.

**2) Class Diagram (max. 2 pts):**

- The method is correctly represented in the class diagram.

If neither the statement number nor the method name is correct, further evaluation for this method will not be conducted.

| Assessment unit                  | Criteria                                                                                                                                                                                                                                                                                                                                        | Code (pts) | Ext. ER Diagram (pts) | Total (pts) |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------|-------------|
| Member function                  | <b><u>Criteria for Evaluating Method Code:</u></b> <ul style="list-style-type: none"> <li>• The method is properly commented.</li> <li>• The method is semantically meaningful and does not contradict the domain description.</li> <li>• The definition matches the description.</li> </ul>                                                    | 5          | 2                     | 7           |
| Member procedure                 |                                                                                                                                                                                                                                                                                                                                                 | 5          | 2                     | 7           |
| MAP or ORDER member function     |                                                                                                                                                                                                                                                                                                                                                 | 5          | 2                     | 7           |
| Static function or procedure     |                                                                                                                                                                                                                                                                                                                                                 | 5          | 2                     | 7           |
| Function returning SYS_REFCURSOR | <ul style="list-style-type: none"> <li>• Successful creation of the method is demonstrated.</li> <li>• Its functionality is convincingly proven with tests.</li> </ul> <b><u>Criteria for Evaluating the Extended ER Diagram:</u></b> <ul style="list-style-type: none"> <li>• The method is correctly represented in the ER diagram</li> </ul> | 4          | 2                     | 6           |

## Functions and Operators

Use of Functions and Operators:

- 1) **Project Passport (max. 2 pts)** – PP includes correct statement numbers where the respective function or operator is demonstrated.
- 2) **Code (max. points see table):** Meaningful and correct use of the function or operator is demonstrated.

| Function or operator | Code: Max. score (pts) |
|----------------------|------------------------|
| VALUE()              | 3                      |
| TABLE()              | 3                      |
| REF()                | 3                      |
| DEREF()              | 3                      |
| TREAT()              | 3                      |

| Function or operator | Code: Max. score (pts) |
|----------------------|------------------------|
| IS OF type           | 3                      |

## Other Constructions

### Type Inheritance (using UNDER):

| Assessment unit         | Criteria                                                                                                                                                                                                                                                         | Max. score (pts) |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Project Passport</b> | <ul style="list-style-type: none"> <li>Correct statement numbers are provided</li> <li>Correct type and subtype names are provided</li> <li><b>If neither the statement number nor the names are correct, further evaluation is not conducted</b></li> </ul>     | <b>1</b>         |
| <b>Code</b>             | <ul style="list-style-type: none"> <li>The code demonstrates the correct creation of the subtype</li> <li>The subtype is meaningful and properly commented</li> <li>Data insertion is demonstrated</li> <li>Data retrieval (detailed) is demonstrated</li> </ul> | <b>3</b>         |
| <b>Ext. ER Diagram</b>  | <ul style="list-style-type: none"> <li>Correct representation in the ER diagram</li> </ul>                                                                                                                                                                       | <b>3</b>         |
| <b>Class Diagram</b>    | <ul style="list-style-type: none"> <li>Correct representation in the class diagram</li> </ul>                                                                                                                                                                    | <b>3</b>         |

### Method Overriding in Inheritance (OVERRIDING):

| Assessment unit         | Criteria                                                                                                                                                                                                                                                                                                                                                                | Max. score (pts) |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Project Passport</b> | <ul style="list-style-type: none"> <li>Correct statement numbers are provided</li> <li>Correct type and method names are provided</li> <li><b>If neither the statement number nor the names are correct, further evaluation is not conducted</b></li> </ul>                                                                                                             | <b>1</b>         |
| <b>Code</b>             | <ul style="list-style-type: none"> <li>The method is properly commented.</li> <li>The method is semantically meaningful and does not contradict the domain description.</li> <li>The definition matches the description (comment).</li> <li>Successful creation of the method is demonstrated.</li> <li>Its functionality is convincingly proven with tests.</li> </ul> | <b>3</b>         |
| <b>Ext. ER Diagram</b>  | <ul style="list-style-type: none"> <li>Correct representation in the ER diagram</li> </ul>                                                                                                                                                                                                                                                                              | <b>3</b>         |
| <b>Class Diagram</b>    | <ul style="list-style-type: none"> <li>Correct representation in the class diagram</li> </ul>                                                                                                                                                                                                                                                                           | <b>3</b>         |

## Grade Reduction

### Grade Reduction:

- 1) For not listing all tools and sources: **up to a complete removal of the grade**
- 2) For significant non-compliance of the report with the template: **up to -100 points**
- 3) For the absence of a group work report or its insufficient quality: **up to -20 points**
- 4) For the video presentation being inaccessible<sup>1</sup> or having an invalid link: **-20 points**
- 5) For the absence of a video presentation: **-20 points**
- 6) For the video presentation's non-compliance with the requirements: **up to -20 points**
- 7) For the absence of conclusions or their poor quality: **up to -10 points**
- 8) If the quality of the Extended ER Diagram is low (but not low enough for it to be completely ungraded) or otherwise does not meet the formatting requirements: **-20 points**
- 9) If the quality of the Class Diagram is low (but not low enough for it to be completely ungraded) or otherwise does not meet the formatting requirements: **-10 points**
- 10) For discrepancies between the content of the code file and the LiveSQL code (if evaluated at all): **-10 points**
- 11) For not using the report template (if the template structure is followed): **-10 points**
- 12) If the code file submitted to ORTUS differs from the code downloaded during evaluation from the LiveSQL link, only in comments or similar details: **-10 points**
- 13) For file names not conforming to the template (per file): **-5 points**
- 14) For other deficiencies: **according to their severity**

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<sup>1</sup> The video presentation is considered inaccessible if it cannot be opened or played due to technical issues, such as an expired, broken, or restricted access link, or if it requires permissions not granted to the evaluator (in case if the video is stored in RTU OneDrive).

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